

Marck Chirza Analysis of Algorithms

① 1.1 $T(n) = 3T(n/2) + n$, $T(1) = 0$, for all n powers of two
1, 2, 4, 8, 16, ...

Let $a_k = T(2^k)$, the recurrence becomes
 $a_k = 3a_{k-1} + 2^k$, $a_0 = 0$

Solving homogeneous $C \cdot 3^k$ + particular $A \cdot 2^k$, with $A = -2$

$$a_k = 2 \cdot 3^k - 2^{k+1}$$

Since

$$n = 2^k \Rightarrow k = \log_2 n$$

$$3^k = n^{\log_2 3}$$

$$\text{So, } T(n) = 2n^{\log_2 3} - 2n$$

Verification.

$$k=0, n=1$$

$$T(1) = 2(1)^{\log_2 3} - 2(1)$$

$$= 2 - 2 = 0 //$$

Induction step

Assume,

$$T(2^k) = 2 \cdot 3^k - 2^{k+1}$$

$$\text{Demonstr, } T(2^{k+1}) = 2 \cdot 3^{k+1} - 2^{k+2}$$

$$T(2^{k+1}) = 3T(2^k) + 2^{k+1}$$

$$= 3(2 \cdot 3^k - 2^{k+1}) + 2^{k+1}$$

$$= 2 \cdot 3^{k+1} - 3 \cdot 2^{k+1} + 2^{k+1}$$

$$= 2 \cdot 3^{k+1} - 2 \cdot 2^{k+1}$$

$$= 2 \cdot 3^{k+1} - 2^{k+2} //$$

1.2 $T(n) = 3T(n-1) + 4T(n-2)$, $T(0) = 0$, $T(1) = 5$

With $T(n) = r^n$

$$r^n = 3r^{n-1} + 4r^{n-2} \Rightarrow \frac{r^n}{r^{n-2}} = \frac{3r^{n-1}}{r^{n-2}} + \frac{4r^{n-2}}{r^{n-2}}$$

$$r^2 - 3r - 4 = 0 \quad (\text{C.E.})$$

$$(r-4)(r+1) = 0 \Rightarrow r_1 = 4, r_2 = -1 \quad (\text{C.R. } r_1 \neq r_2)$$

$$T(n) = C_1(4)^n + C_2(-1)^n$$

$$T(0) = C_1 + C_2 = 0$$

$$T(1) = 4C_1 - C_2 = 5$$

$$5C_1 = 5 \Rightarrow C_1 = 1, C_2 = -1$$

$$T(n) = 4^n - (-1)^n$$

Verification

$$T(0) = 1 - 1 = 0, \quad T(1) = 4 - (-1) = 5$$

Induction step

Assume true for $n=k$ and $n=k-2$

$$T(k) = 3T(k-1) + 4T(k-2)$$

$$= 3(4^{k-1} - (-1)^{k-1}) + 4(4^{k-2} - (-1)^{k-2})$$

$$= 4^{k-2}(3 \cdot 4 + 4) - (-1)^{k-2}(3(-1) + 4)$$

$$= 4^k - (-1)^{k-2}$$

$$= 4^k - (-1)^k \quad \checkmark$$

$$\underline{1.3} \quad T(n) = 5T(n-1) - 6T(n-2)$$

$$\underline{1.3} \quad T(n) = 5T(n-1) - 6T(n-2), \quad T(0)=0, \quad T(1)=1$$

$$r^2 - 5r + 6 = 0 \quad (\text{C.E.})$$

$$(r-3)(r-2) = 0 \Rightarrow r_1 = 3, r_2 = 2 \quad (\text{C.R., } r_1 \neq r_2)$$

$$T(n) = C_1(3)^n + C_2(2)^n$$

$$T(0) = C_1 + C_2 = 0$$

$$T(1) = 3C_1 + 2C_2 = 1$$

$$\text{From the first } C_1 = -C_2$$

Substituting

$$-3C_2 + 2C_1 = 1$$

$$C_2 = -1, C_1 = 1$$

$$T(n) = 3^n - 2^n //$$

Verifications

$$T(0) = 1 - 1 = 0, \quad T(1) = 3 - 2 = 1$$

Induction step

Assume true for $k-1, k-2$

$$T(k) = 5(3^{k-1} - 2^{k-1}) - 6(3^{k-2} - 2^{k-2})$$

$$= 3^{k-2}(15-6) - 2^{k-2}(10-6)$$

$$= 9 \cdot 3^{k-2} - 4 \cdot 2^{k-2}$$

$$= 3^k - 2^k //$$

$$1.4 // x(n) = 2x(n-1) + 4x(n-2), \quad x(0)=1, \quad x(1)=2, \quad \text{for } n \geq 1$$

$$r^2 - 2r - 4 = 0 \quad (1.6)$$

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r = \frac{2 \pm \sqrt{4+16}}{2} = 1 \pm \sqrt{5}$$

$$r_1 = 1 + \sqrt{5}, \quad r_2 = 1 - \sqrt{5} \quad (\text{C.R. } r_1 \neq r_2)$$

$$x(n) = c_1 (1 + \sqrt{5})^n + c_2 (1 - \sqrt{5})^n$$

$$x(0) = c_1 + c_2 = 1$$

$$x(1) = c_1 (1 + \sqrt{5}) + c_2 (1 - \sqrt{5}) = 2$$

$$\Rightarrow (c_1 + c_2) + \sqrt{5}(c_1 - c_2) = 2$$

$$\Rightarrow c_1 - c_2 = \frac{1}{\sqrt{5}}$$

Solving the 2×2 system

$$c_1 = \frac{\sqrt{5}+1}{2\sqrt{5}}, \quad c_2 = \frac{\sqrt{5}-1}{2\sqrt{5}}$$

$$x(n) = \frac{\sqrt{5}+1}{2\sqrt{5}} (1+\sqrt{5})^n + \frac{\sqrt{5}-1}{2\sqrt{5}} (1-\sqrt{5})^n //$$

Verification

$$x(0) = c_1 + c_2 = 1$$

$$x(1) = 2$$

$$c_1 = \frac{\sqrt{5}+1}{2\sqrt{5}}, \quad c_2 = \frac{\sqrt{5}-1}{2\sqrt{5}} \quad \text{y } r_1 = 1 + \sqrt{5}, \quad r_2 = 1 - \sqrt{5}$$

We want to prove, $x(1) = c_1 r_1 + c_2 r_2 = 2$

$$c_1 r_1 = \frac{\sqrt{5}+1}{2\sqrt{5}} (1+\sqrt{5}) = \frac{(1+\sqrt{5})(1+\sqrt{5})}{2\sqrt{5}} = \frac{(1+\sqrt{5})^2}{2\sqrt{5}}$$

$$c_1 r_1 = \frac{6+2\sqrt{5}}{2\sqrt{5}} = \frac{3+\sqrt{5}}{\sqrt{5}}$$

Now, $C_2 r_2$

$$C_2 r_2 = \frac{\sqrt{5}-1}{2\sqrt{5}} (1-\sqrt{5})$$

Como $(1-\sqrt{5}) = (\sqrt{5}-1)$ so,

$$C_2 r_2 = \frac{\sqrt{5}-1}{2\sqrt{5}} \cdot (- (\sqrt{5}-1))$$

$$= - \frac{(\sqrt{5}-1)^2}{2\sqrt{5}}$$

$$C_2 r_2 = - \frac{6-2\sqrt{5}}{2\sqrt{5}} = - \frac{6+2\sqrt{5}}{2\sqrt{5}} = \frac{-3+\sqrt{5}}{\sqrt{5}}$$

Now, $x(1) = C_1 r_1 + C_2 r_2$

$$C_1 r_1 + C_2 r_2 = \frac{3+\sqrt{5}}{\sqrt{5}} + \frac{-3+\sqrt{5}}{\sqrt{5}}$$

$$= \frac{(3+\sqrt{5}) + (-3+\sqrt{5})}{\sqrt{5}}$$

$$= \frac{2\sqrt{5}}{\sqrt{5}} = 2 //$$

$$x(1) = 2 //$$

Induction step.

Assume true for $n=k-1$ and $n=k-2$

$$x(k-1) = C_1 r_1^{k-1} + C_2 r_2^{k-1}$$

$$x(k-2) = C_1 r_1^{k-2} + C_2 r_2^{k-2}$$

$$x(k) = C_1 r_1^k + C_2 r_2^k$$

$$x(k) = 2x(k-1) + 4x(k-2)$$

$$x(k) = 2(C_1 r_1^{k-1} + C_2 r_2^{k-1}) + 4(C_1 r_1^{k-2} + C_2 r_2^{k-2})$$

$$x(k) = C_1 (2r_1^{k-1} + 4r_1^{k-2}) + C_2 (2r_2^{k-1} + 4r_2^{k-2})$$

$$x(k) = C_1 r_1^{k-2} (2r_1 + 4) + C_2 r_2^{k-2} (2r_2 + 4)$$

we have $r^2 - 2r - 4 = 0$

$$r^2 = 2r + 4$$

Substituting

$$2r_1 + 4 = r_1^2, \quad 2r_2 + 4 = r_2^2$$

$$x(k) = C_1 r_1^{k-2} \cdot v_1^2 + C_2 r_2^{k-2} \cdot v_2^2$$

$$x(k) = C_1 r_1^k + C_2 r_2^k \checkmark \checkmark$$